
Lecture 8: Latch and Flip Flop Design

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Outline

- Recent interest in latches and flip-flops
- Timing and Power metrics
- Design and optimization tradeoffs
- Master-slave vs. Pulse-triggered Latch
- Representative designs
- Comparison

Recent Interest in Flip-Flops

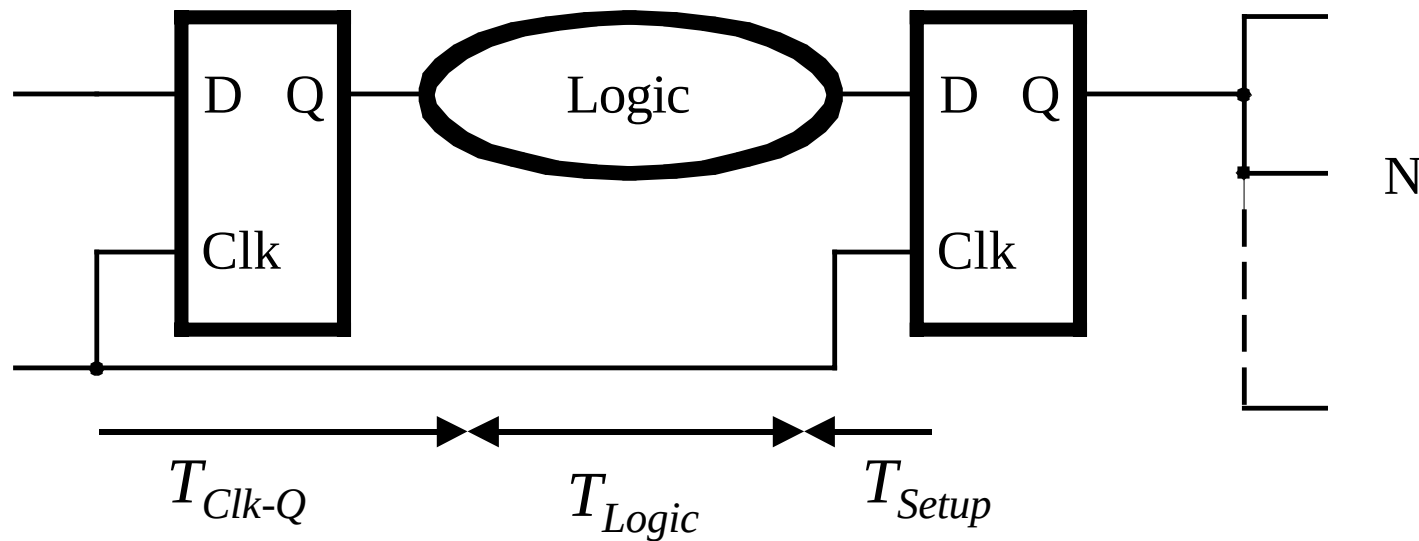
- Trends in high-performance systems
 - Higher clock frequency
 - More transistors on chip
- Consequences
 - Increased flip-flop overhead relative to cycle time
 - Cycle time 10 - 20 FO4 delays, flop overhead 2 - 4 FO4
 - Difficult to control both edges of the clock
 - Higher impact of clock skew
 - Higher crosstalk and substrate coupling
 - Higher power consumption
 - expensive packages and cooling systems
 - limit in performance
 - Clock burns up to 40%, flops up to 20% of total power

Requirements in the Flip-Flop Design

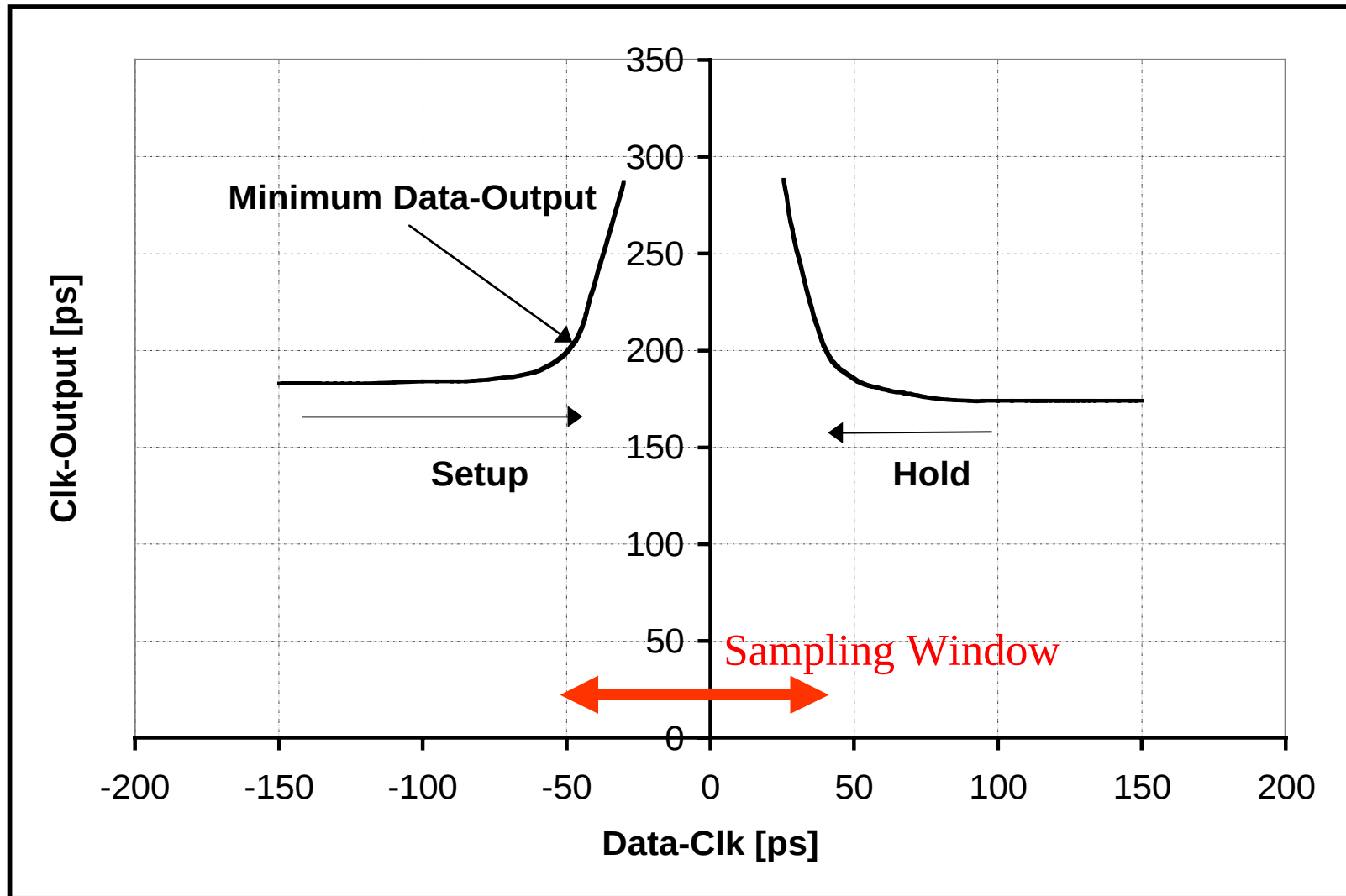
- Small Clk-Output delay, Narrow sampling window
- Low power
- Small clock load
- High driving capability (increased levels of parallelism)
 - Typical flip-flop load in a 0.18 μ m CMOS ranges from 50fF to over 200fF, with typical values of 100-150fF in critical paths
- Integration of logic into the flop
- Multiplexed or clock scan
- Crosstalk insensitivity
 - dynamic/high impedance nodes are affected

Flip-Flop Delay

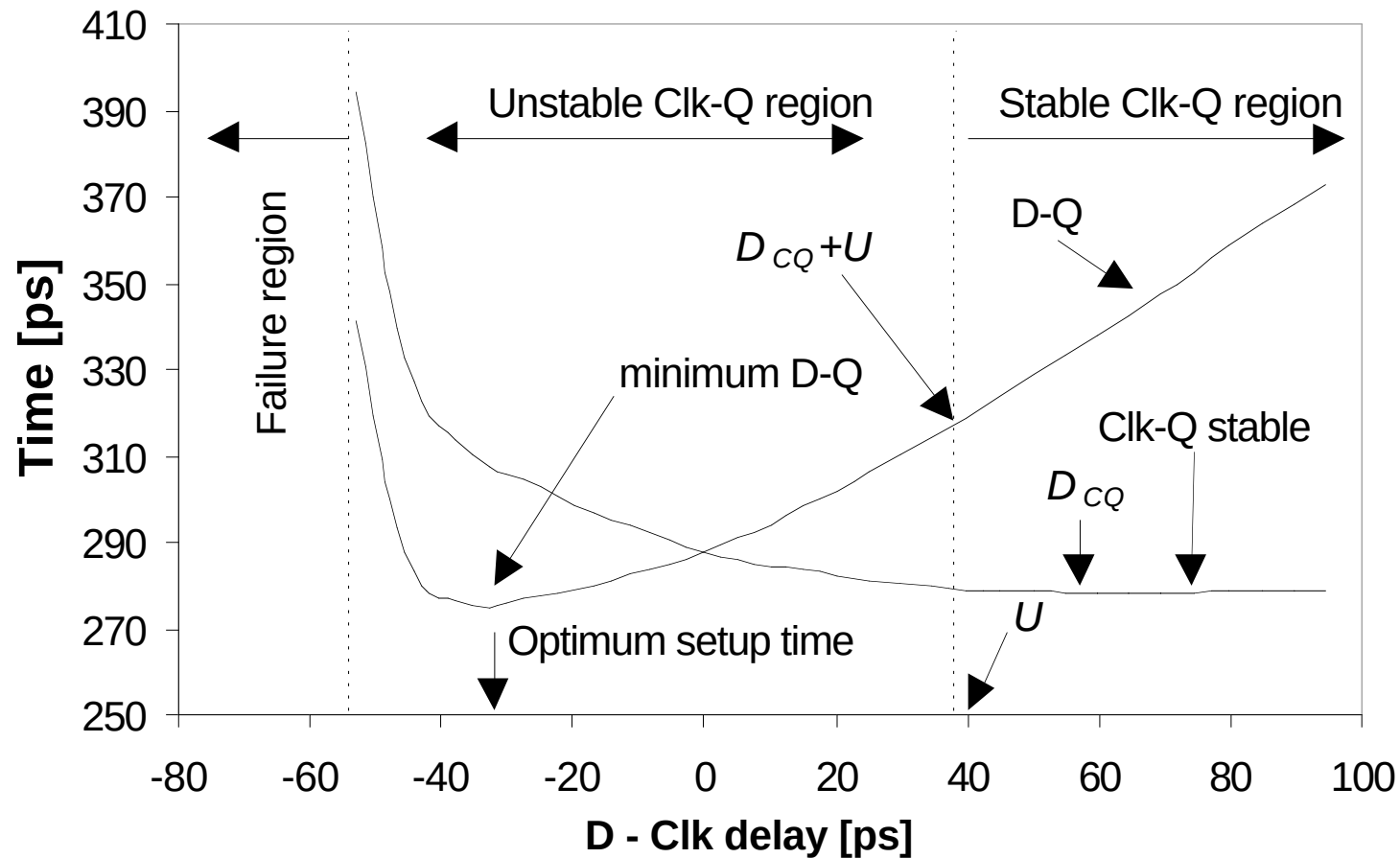
- Sum of setup time and Clk-output delay is the only true measure of the performance with respect to the system speed
- $T = T_{Clk-Q} + T_{Logic} + T_{setup} + T_{skew}$



Delay vs. Setup/Hold Times



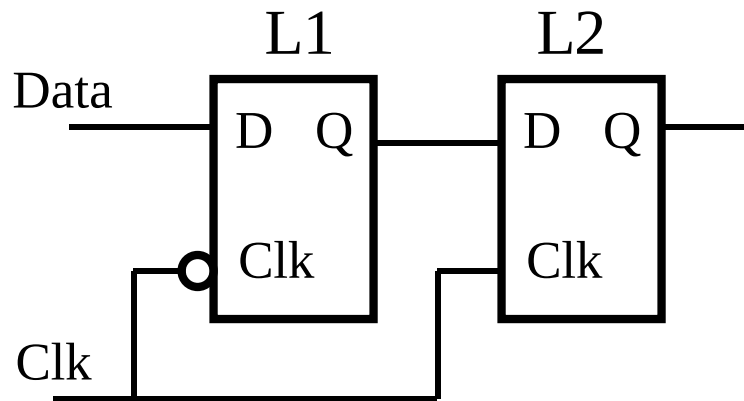
Timing parameters, details



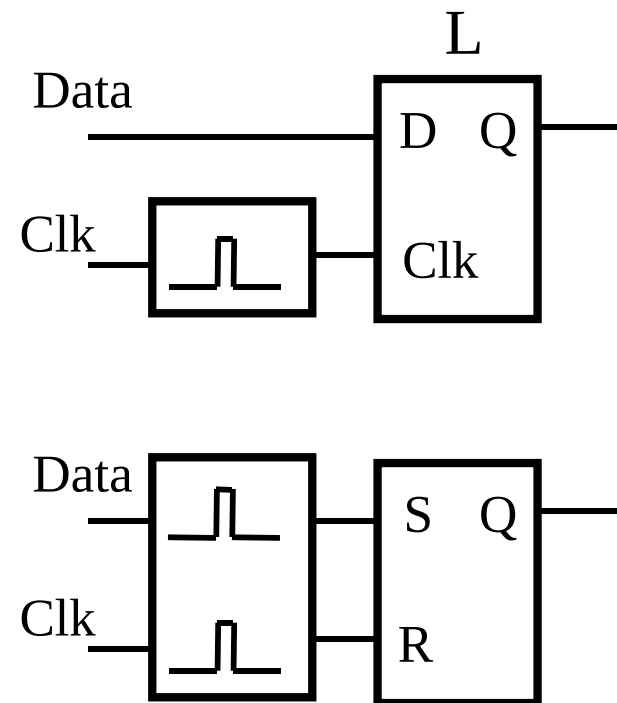
The best point to pick on delay curve is **minimum D-Q**

Types of State-Elements

Master-Slave Latch



Pulse-Triggered Latch

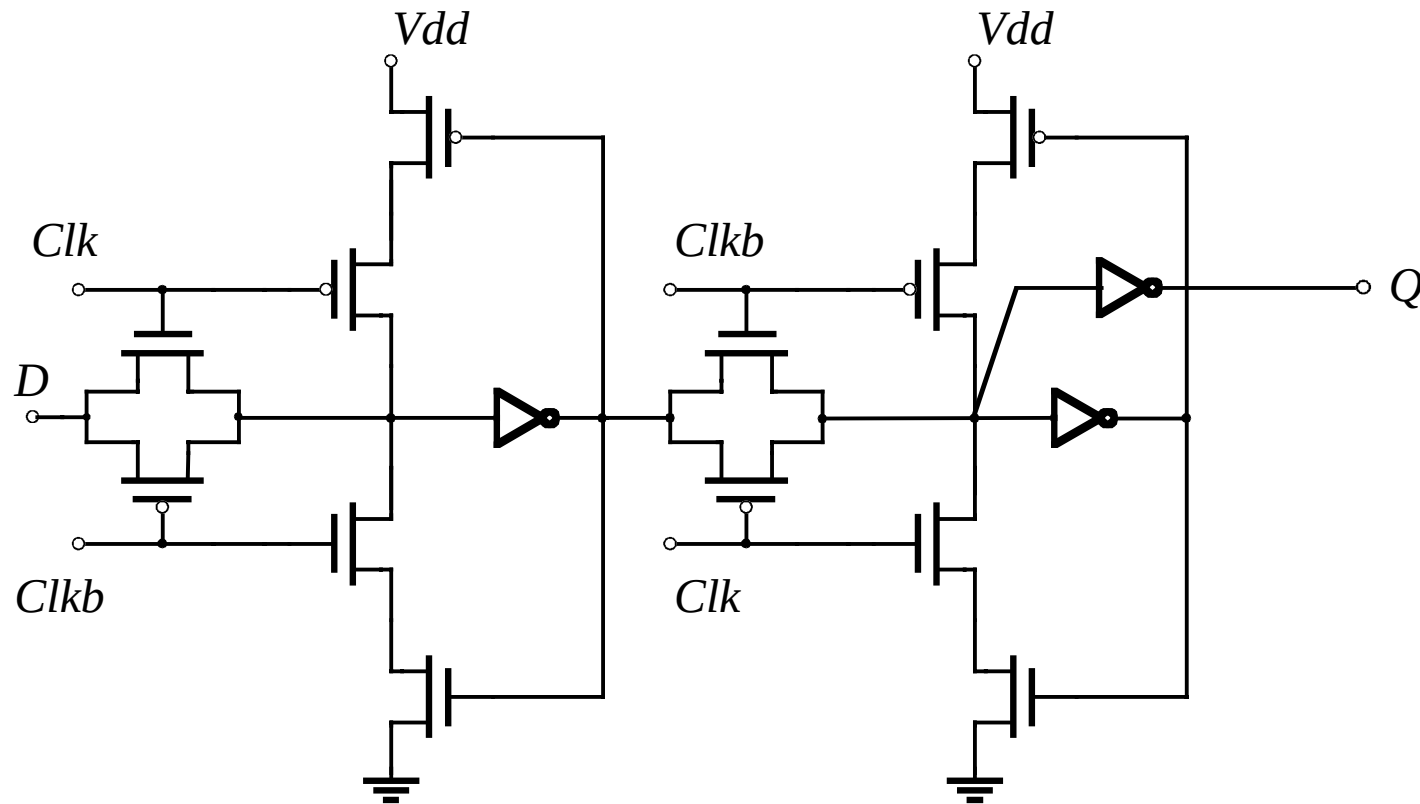


Master-Slave Latches

- Positive setup times
- Two clock phases:
 - distributed globally
 - generated locally
- Small penalty in delay for incorporating MUX
- Some circuit tricks needed to reduce the overall delay

T-G Master-Slave Latch

- PowerPC 603 (Gerosa, JSSC 12/94)

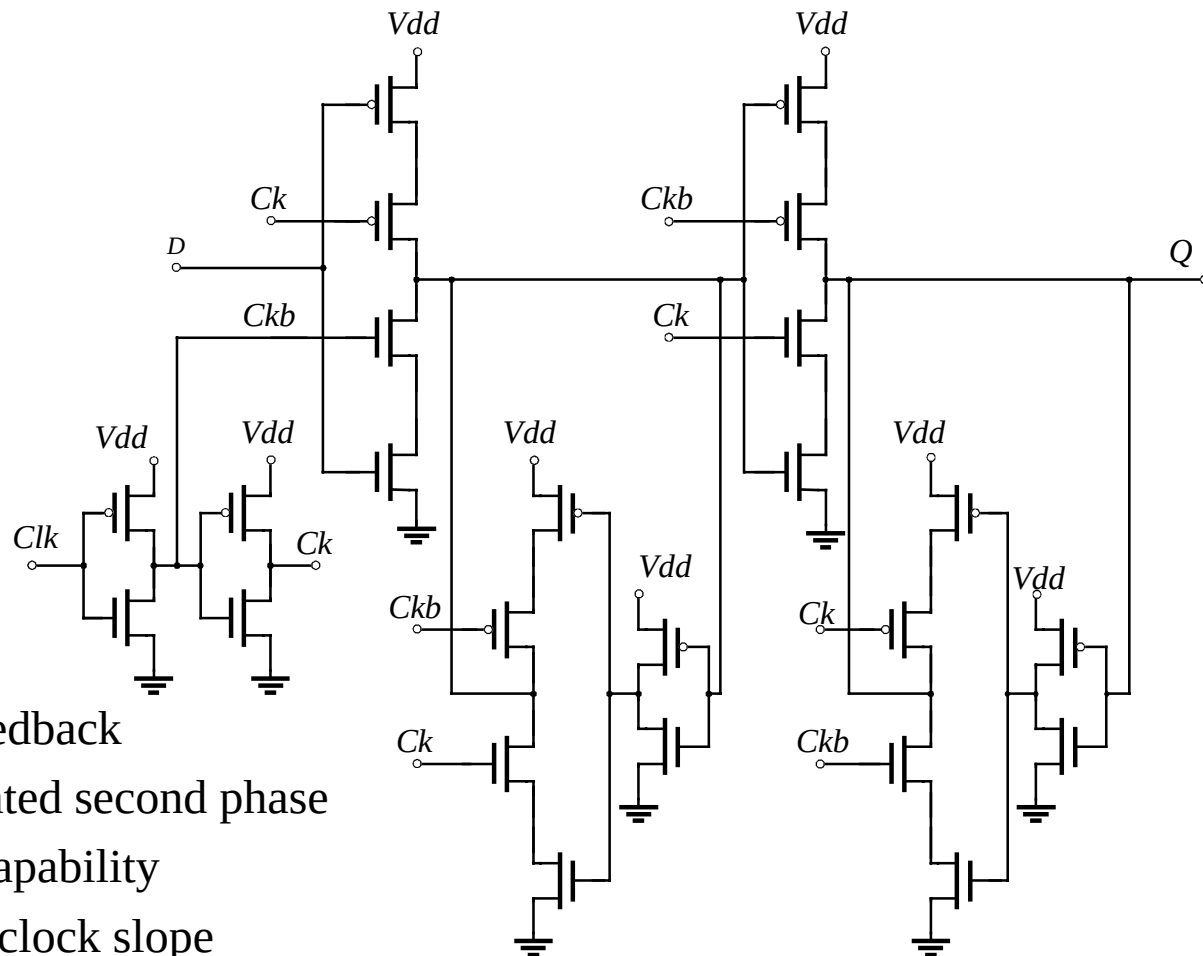


T-G Master-Slave Latch

- Low power feedback
- Unbuffered input
 - input capacitance depends on the phase of the clock
 - over-shoot and under-shoot with long routes
 - wirelength must be restricted at the input
- Clock load is high
- Low power
- Small clk-output delay, but positive setup
- Easily embedded scan or mux

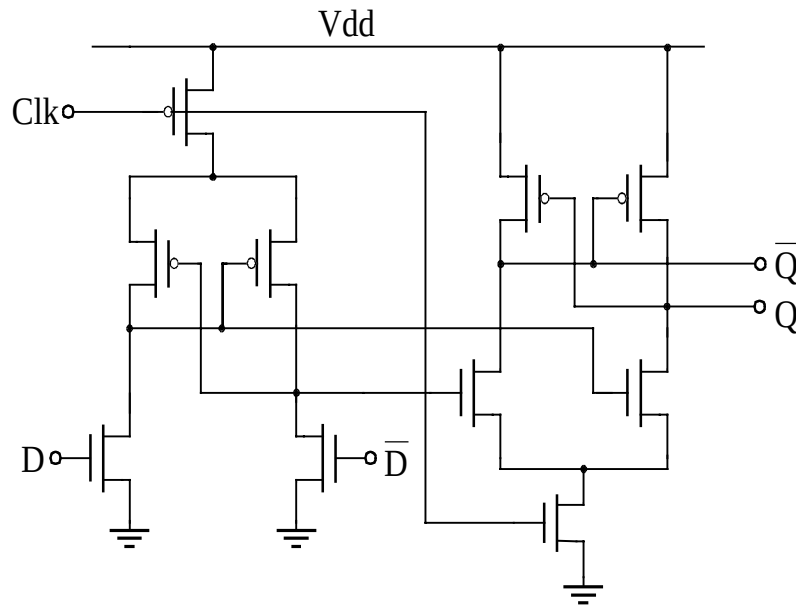
C²MOS MS Latches

Y. Suzuki, "Clocked CMOS Calculator Circuitry", *IEEE J. Solid-State Circuits*, Dec. 1973

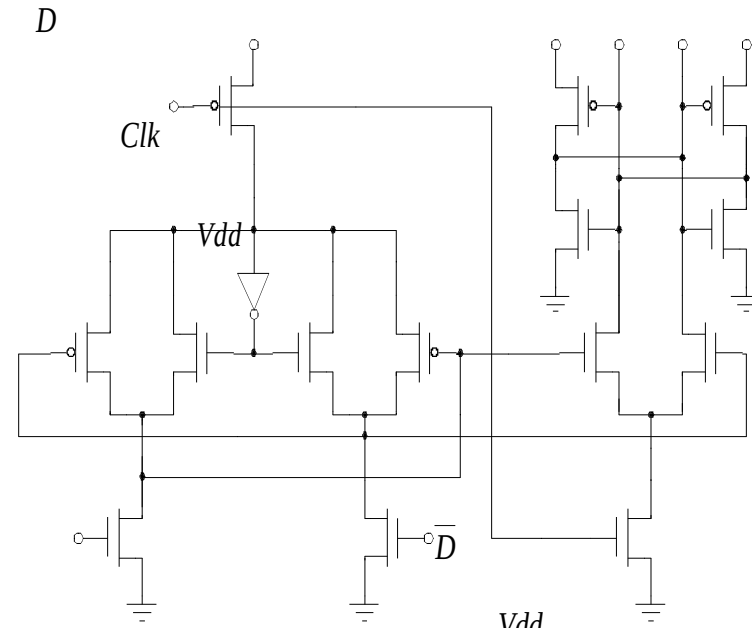


- Low power feedback
- Locally generated second phase
- Poor driving capability
- Robustness to clock slope

Single-Transistor-Clocked MS latches



DSTC



SSTC

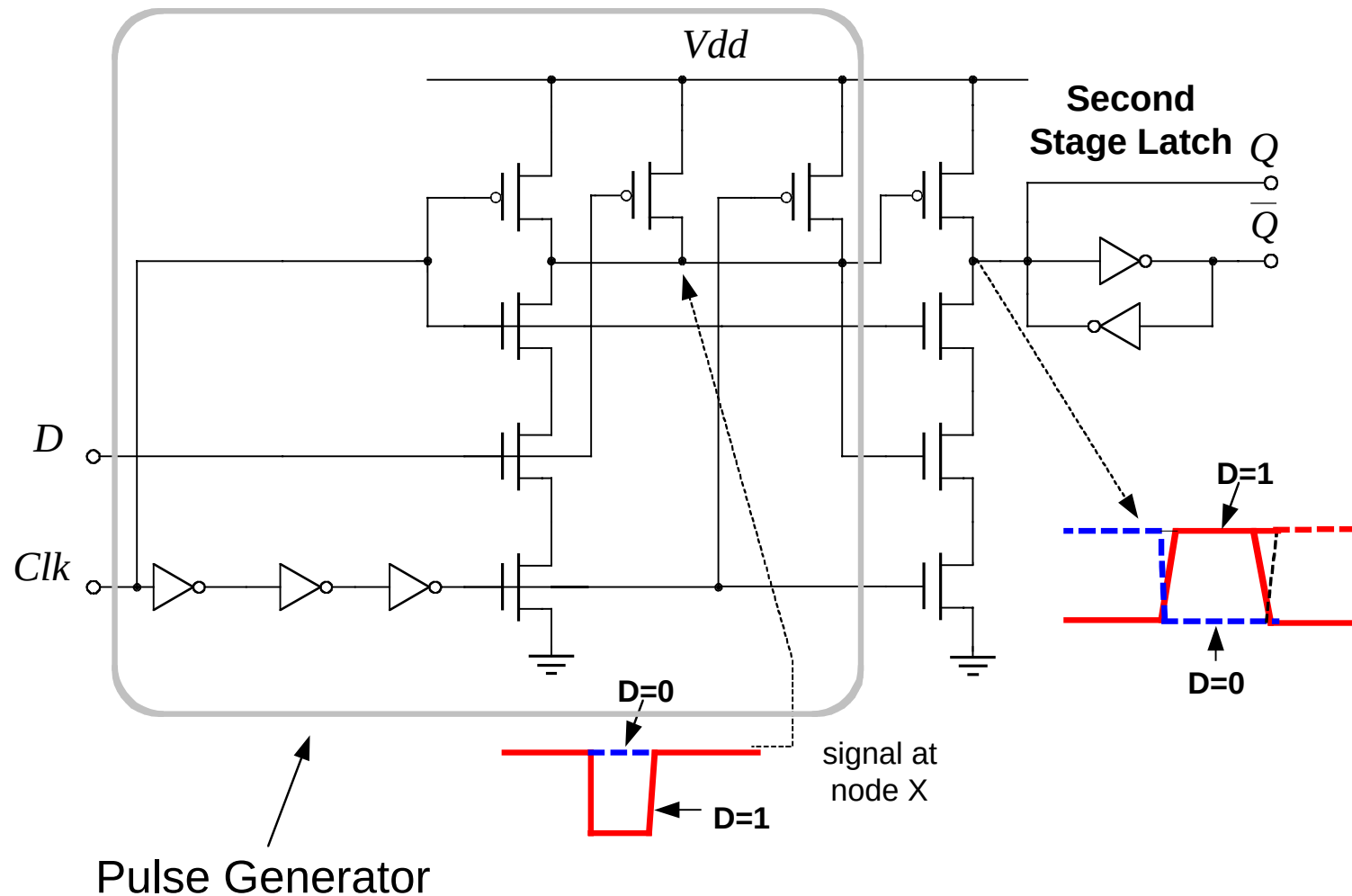
- Yuan and Svensson, JSSC Jan. '97
- Ratioed DCVS and SRPL based designs
- Relatively small clock load
- Very sensitive to input glitching
- Capacitive coupling and charge sharing related speed and power problems

Pulse-Triggered Latches

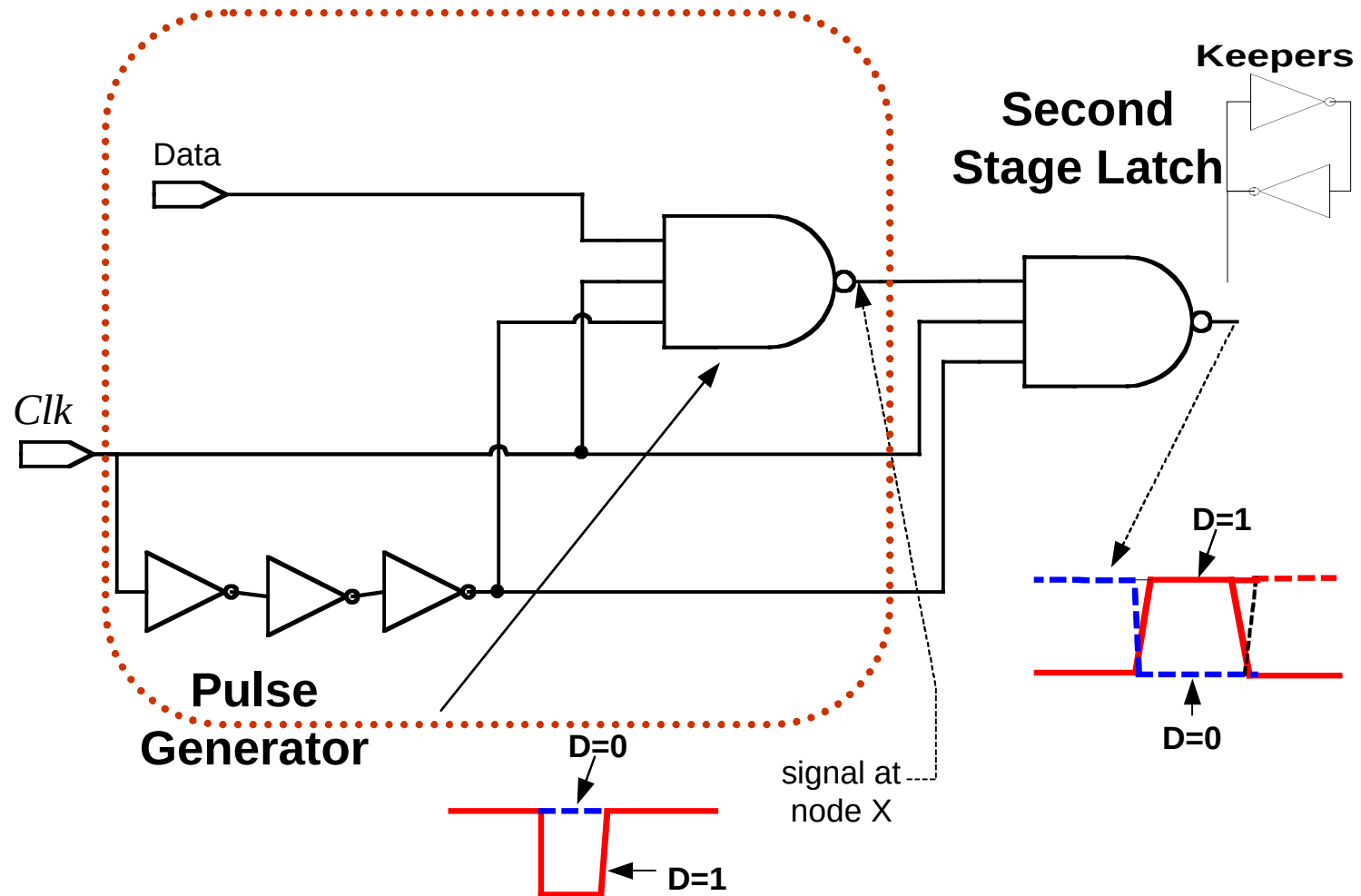
- First stage is a pulse generator
 - generates a pulse (glitch) on a rising edge of the clock
- Second stage is a latch
 - captures the pulse generated in the first stage
- Pulse generation results in a negative setup time
- Frequently exhibit a soft edge property
- Must check for hold time violations

Note: power is always consumed in the clocked pulse generator

Hybrid Latch Flip-Flop (H. Partovi, ISSCC'96)

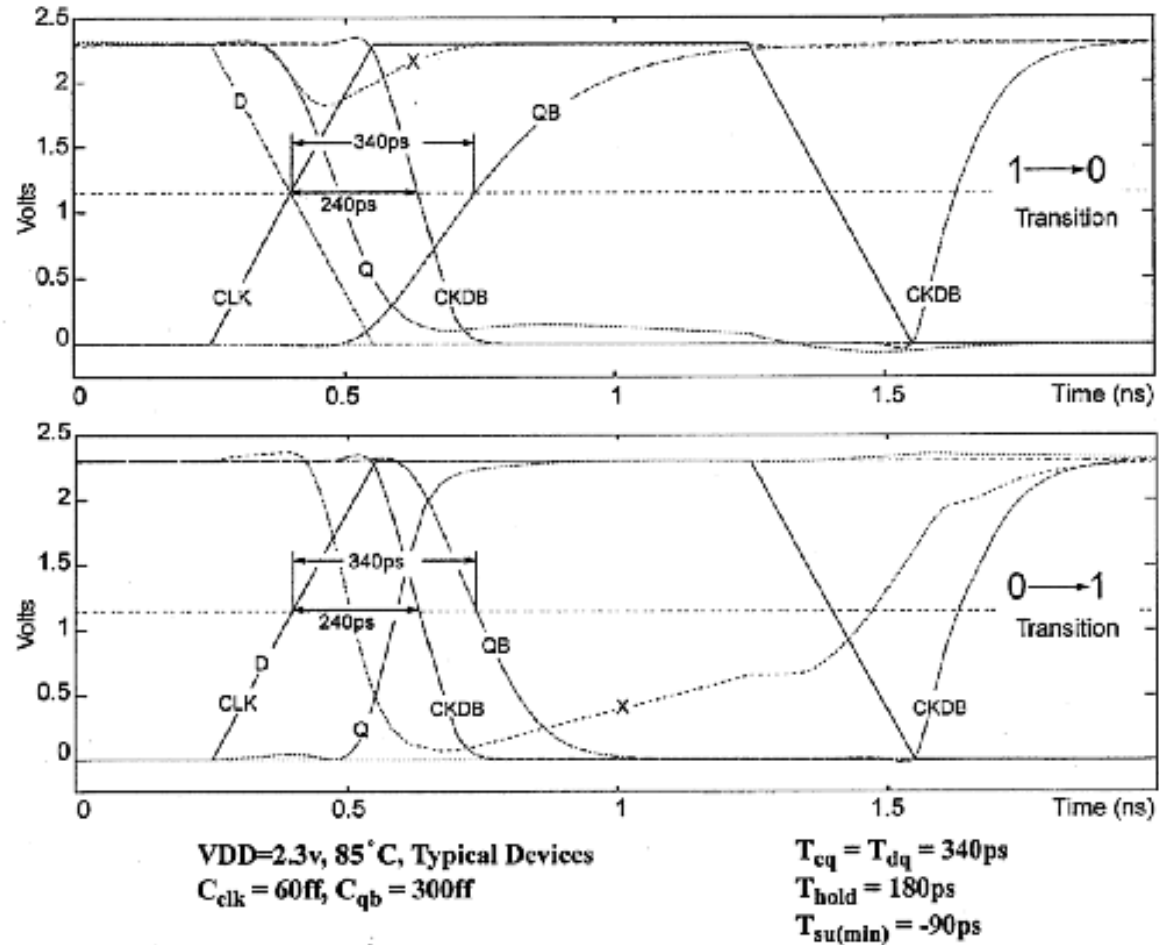


HLFF – pulse generation



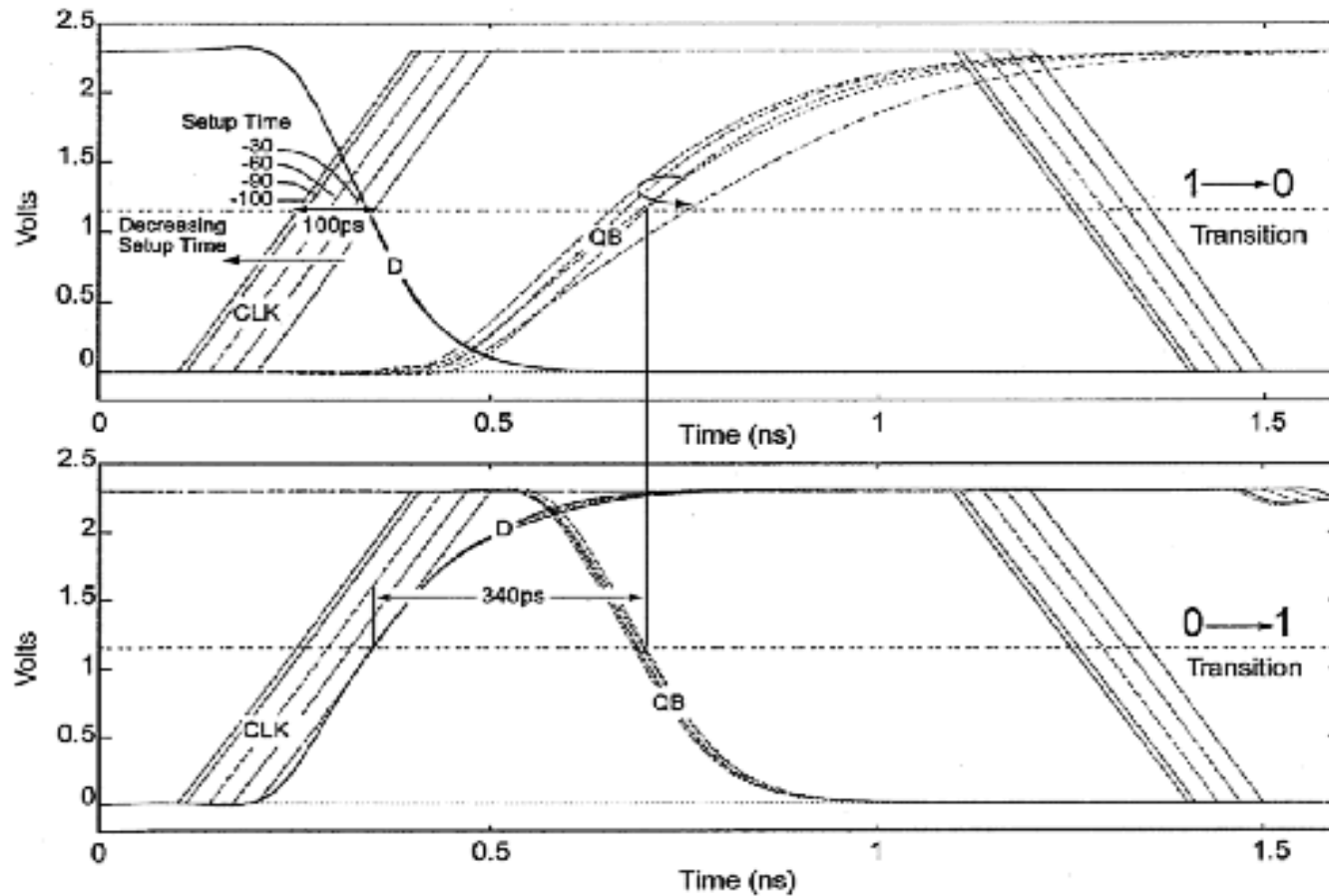
HLFF Operation

- 1-0 and 0-1 transitions at the input with 0ps setup time



Hybrid Latch Flip-Flop

Skew absorption



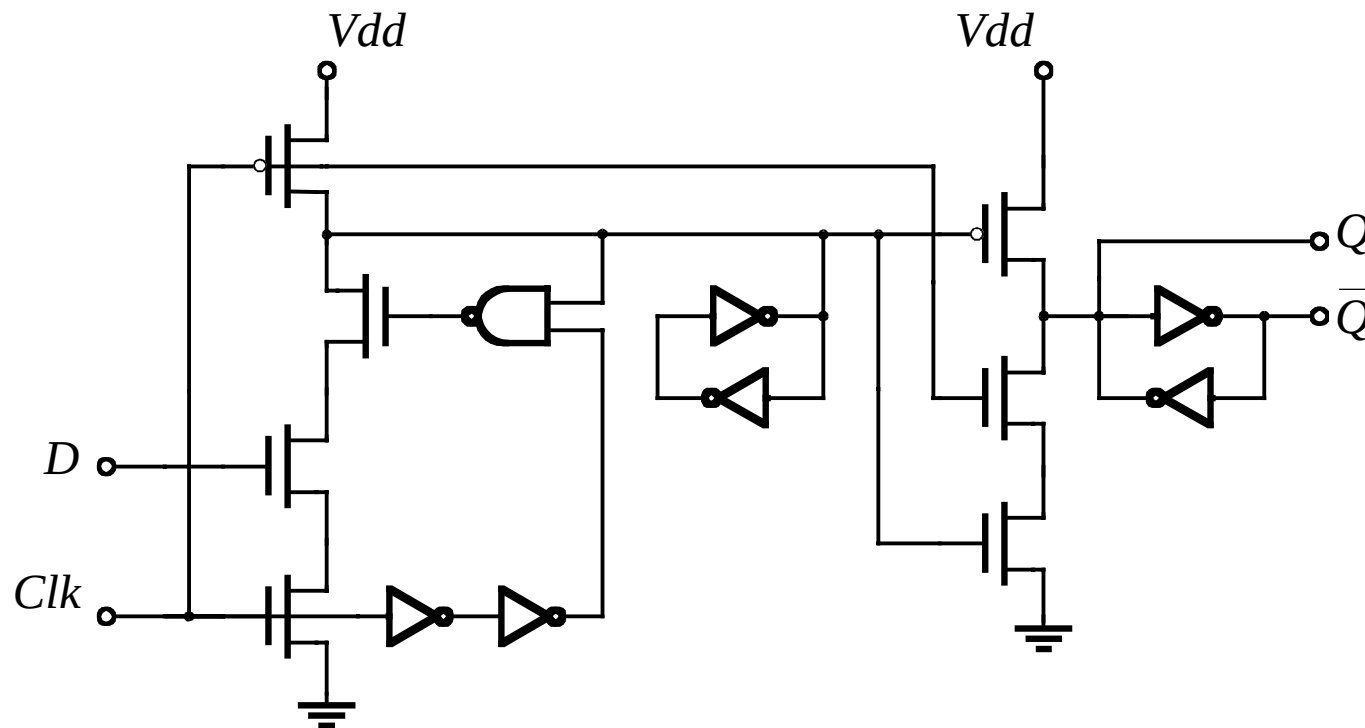
Partovi *et al*, ISSCC'96

Hybrid Latch Flip-Flop

- Flip-flop features:
 - single phase clock
 - edge triggered, on one clock edge
- Latch features: Soft clock edge property
 - brief transparency, equal to 3 inverter delays
 - negative setup time
 - allows slack passing
 - absorbs skew
- Hold time is comparable to HLFF delay
 - minimum delay between flip-flops must be controlled
- Fully static
- Possible to incorporate logic

Semi-Dynamic Flip-Flop (SDFF)

- Sun UltraSparc III, Klass, VLSI Circuits'98



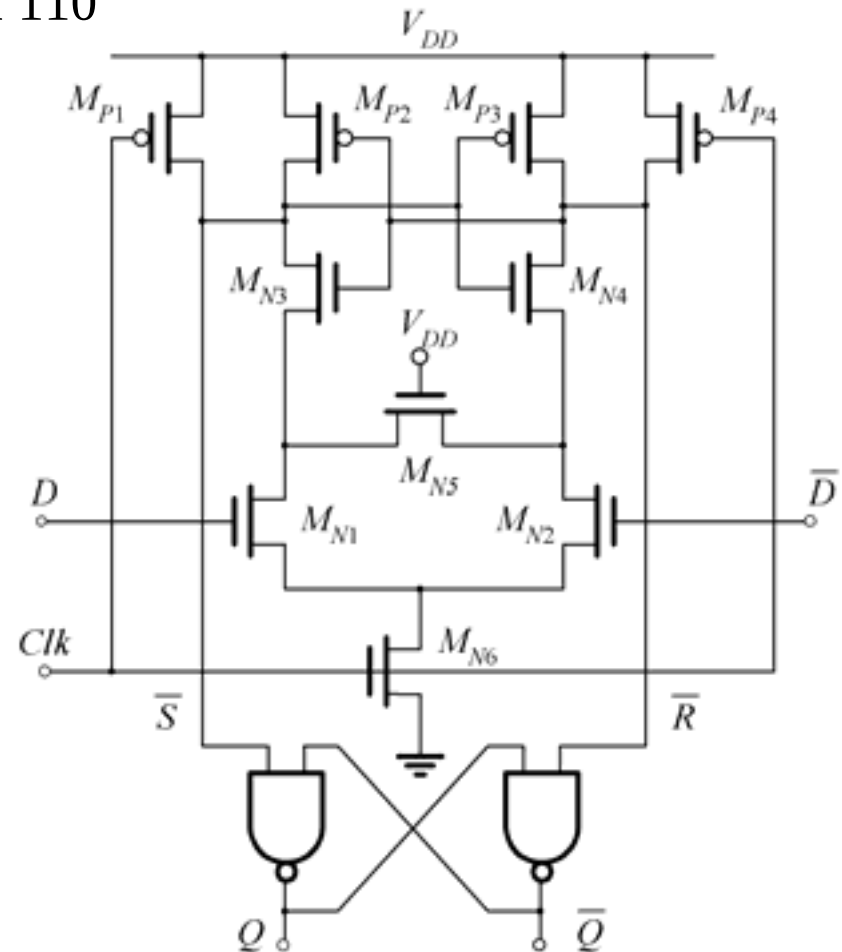
- Soft edge conditioned by data since first stage is precharged - cross-coupled latch is added for robustness
- Small penalty for adding logic
- Latch has one transistor less in stack - faster than HLFF, but 1-1 glitch exists

Sense-amplifier-based flip-flop

Madden & Bowhill, 1990, Matsui *et al.* 1994.

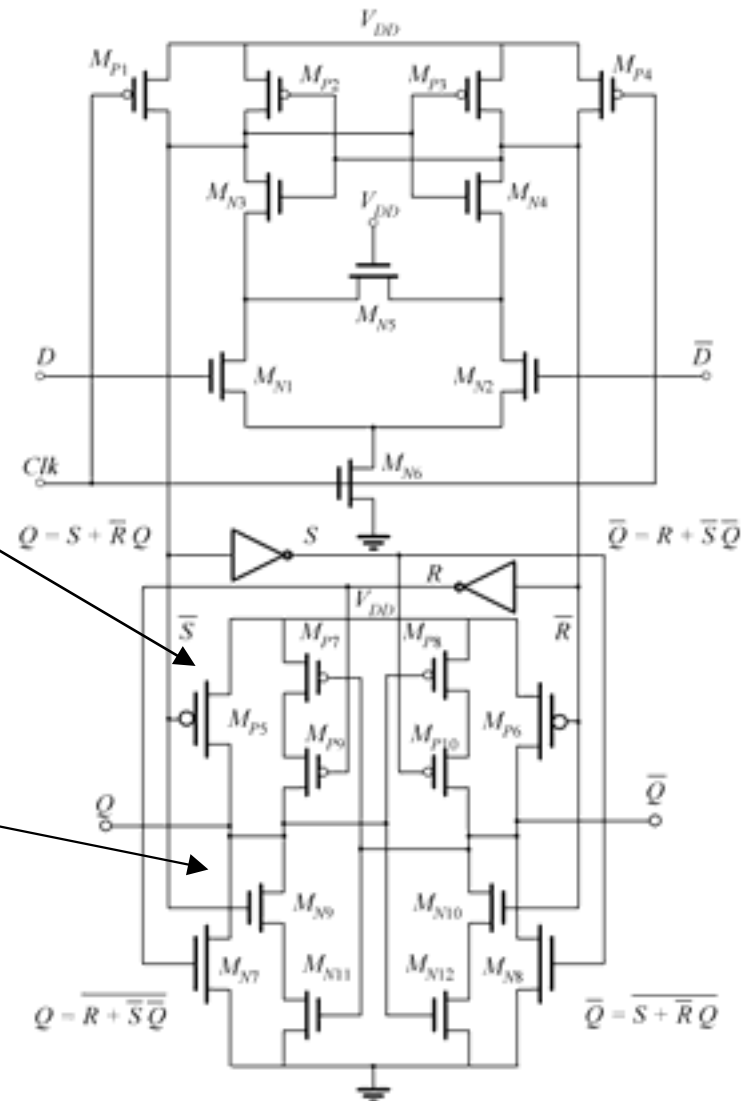
DEC Alpha 21264, StrongARM 110

- First stage is a sense amplifier
- On rising clock edge monotonic S_b or R_b trigger the S-R latch
- Cross-coupled NAND - speed bottleneck
- Big power savings in reduced swing designs
- Nice interface to/from domino logic



Modified Sense Amplifier-Based Flip-Flop

- The first stage is unchanged sense amplifier
- Second stage is sized to provide maximum switching speed
- Driver transistors are large
- Keeper transistors are small and disengaged during transitions



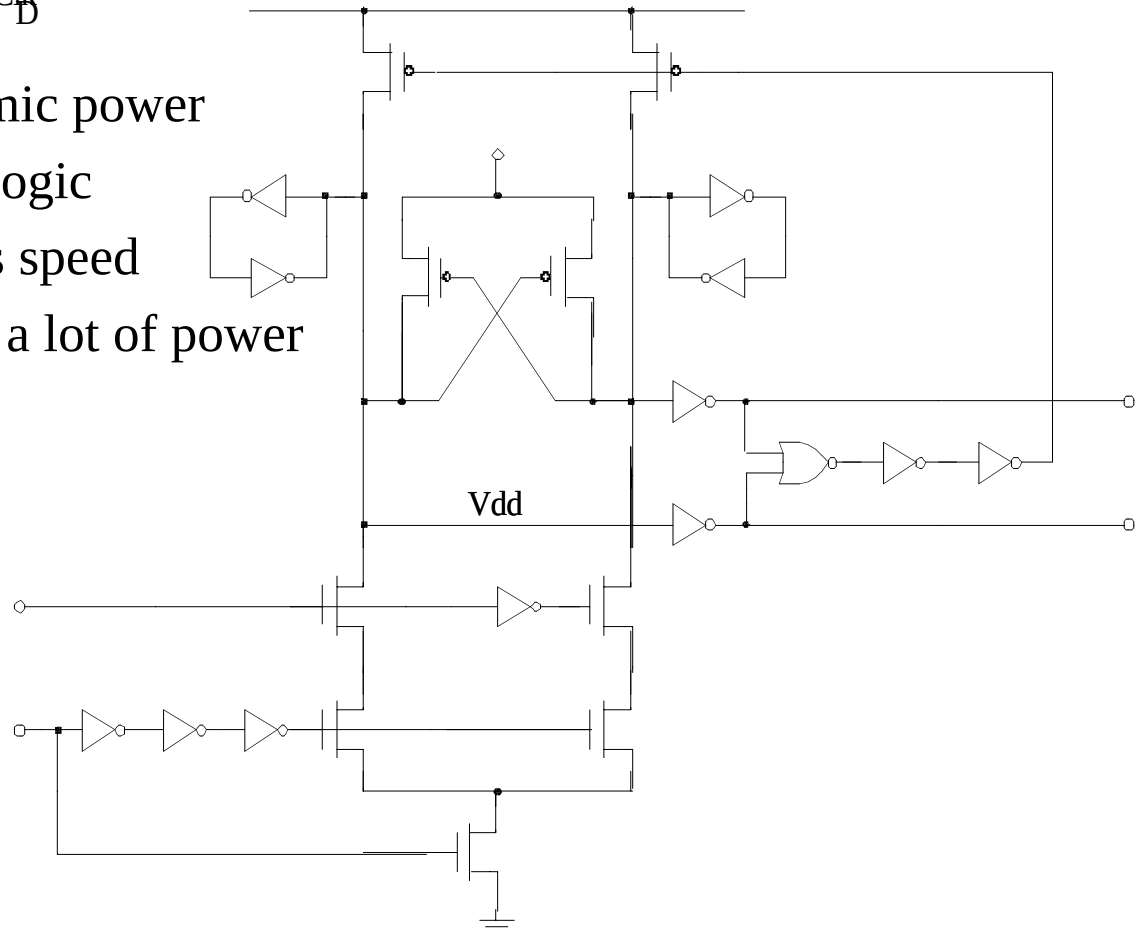
Nikolic & Stojanovic, ISSCC '99

Modified Sense Amplifier-Based Flip-Flop

- Delay of each of the outputs is independent of the load on the other output
- Delay of Q and $\bar{Q}$ is symmetrical as opposed to the NAND based design
- Convenient for dual rail logic and driving strength for standard CMOS is effectively doubled
- SAFF presents a small clock load, small setup time and all the advantages of original design
- Possible tradeoff between speed and robustness to cross-talk

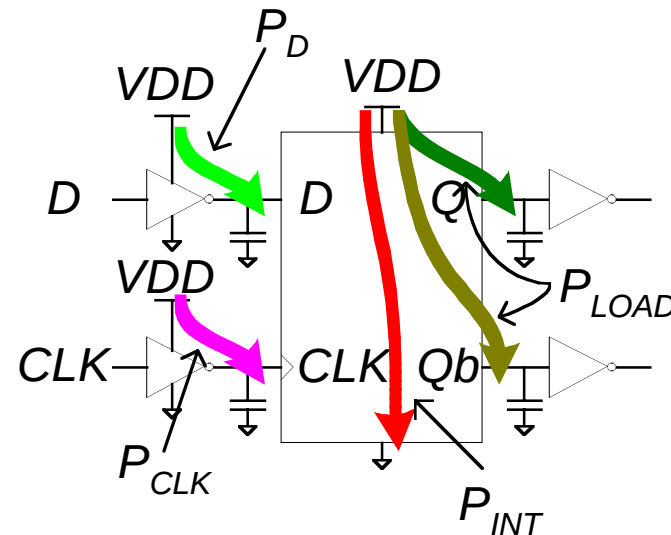
K-6 Dual-Rail ETL

- Self-reset property^{clk}
 - increases dynamic power
 - drives domino logic
- Precharge increases speed
- Very fast but burns a lot of power
- Small clock load



Power and Delay Definitions

- All power related to the SE can be divided into:
 - Input power
 - Data power (P_D)
 - Clock power (P_{CLK})
 - Internal power (P_{INT})
 - Load power (P_{LOAD})
- P_{LOAD} can be merged into P_{INT}
- Internal power is a function of
 - data activity ratio (α) – number of captured data transitions with respect to number of clock transitions ($\alpha_{max}=100\%$)
 - no activity (0000... and 1111...)
 - maximum activity (0101010..)
 - average activity (random sequence)
 - Glitching activity



$$P_{tot} = P_{internal} + \sum_{inputs(D,CLK)} P_{driver}$$

Delay is (minimum $D-Q$)
 $Clk-Q$ + setup time

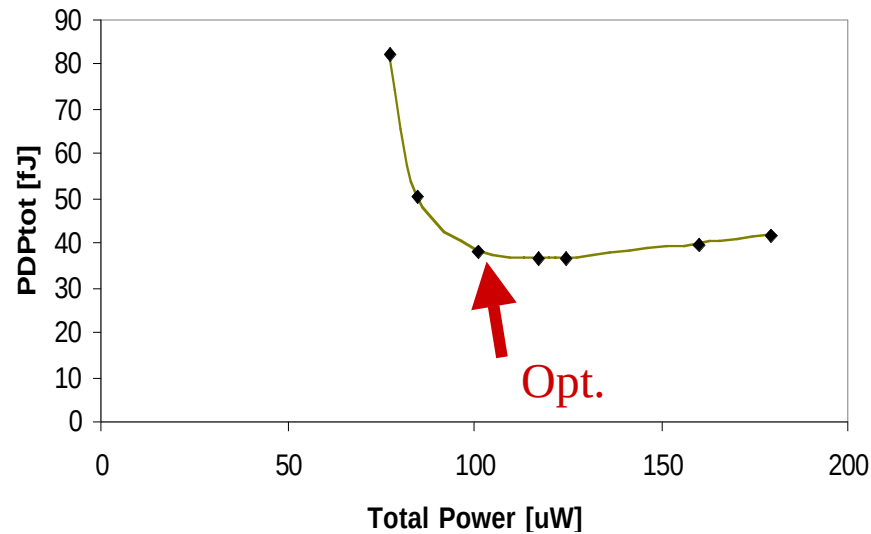
State Element Performance Metrics

It is always possible trade power for speed

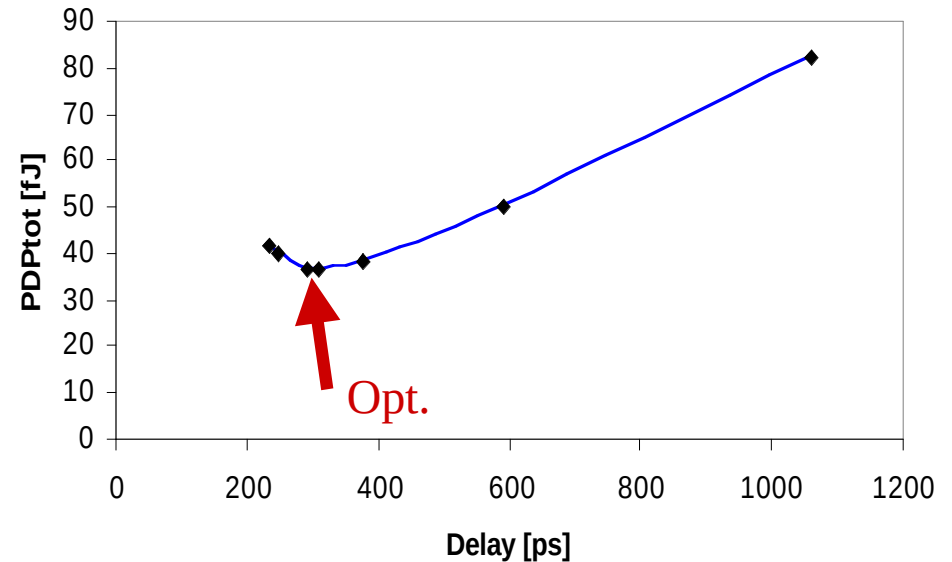
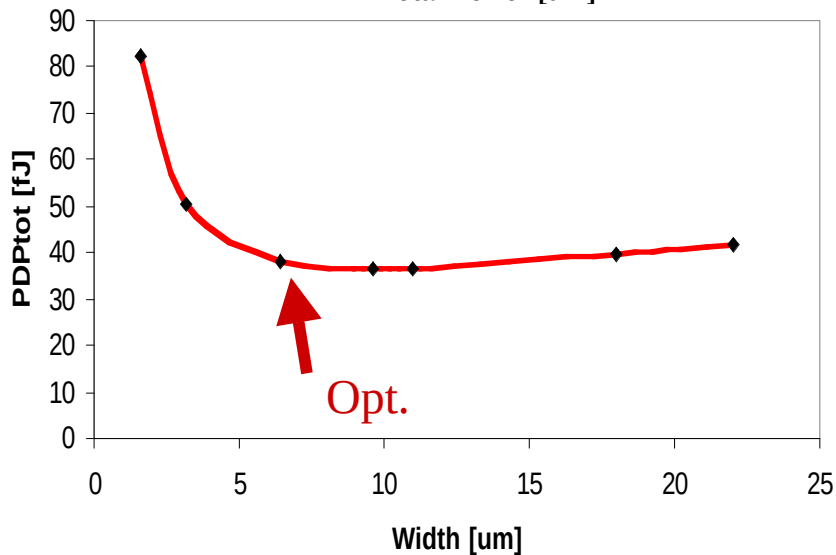
Common metrics:

- **Power-Delay Product (PDP)**
 - Misleading measure
 - Good only if measured at constant frequency = EDP
- **EDP - Energy-Delay Product (EDP)**
 - More accurate measure (Gonzalez & Horowitz)
- **ED²P – Energy-Delay²-Product**
 - A new measure, being justified by new results (Hofstee, Nowka, IBM)

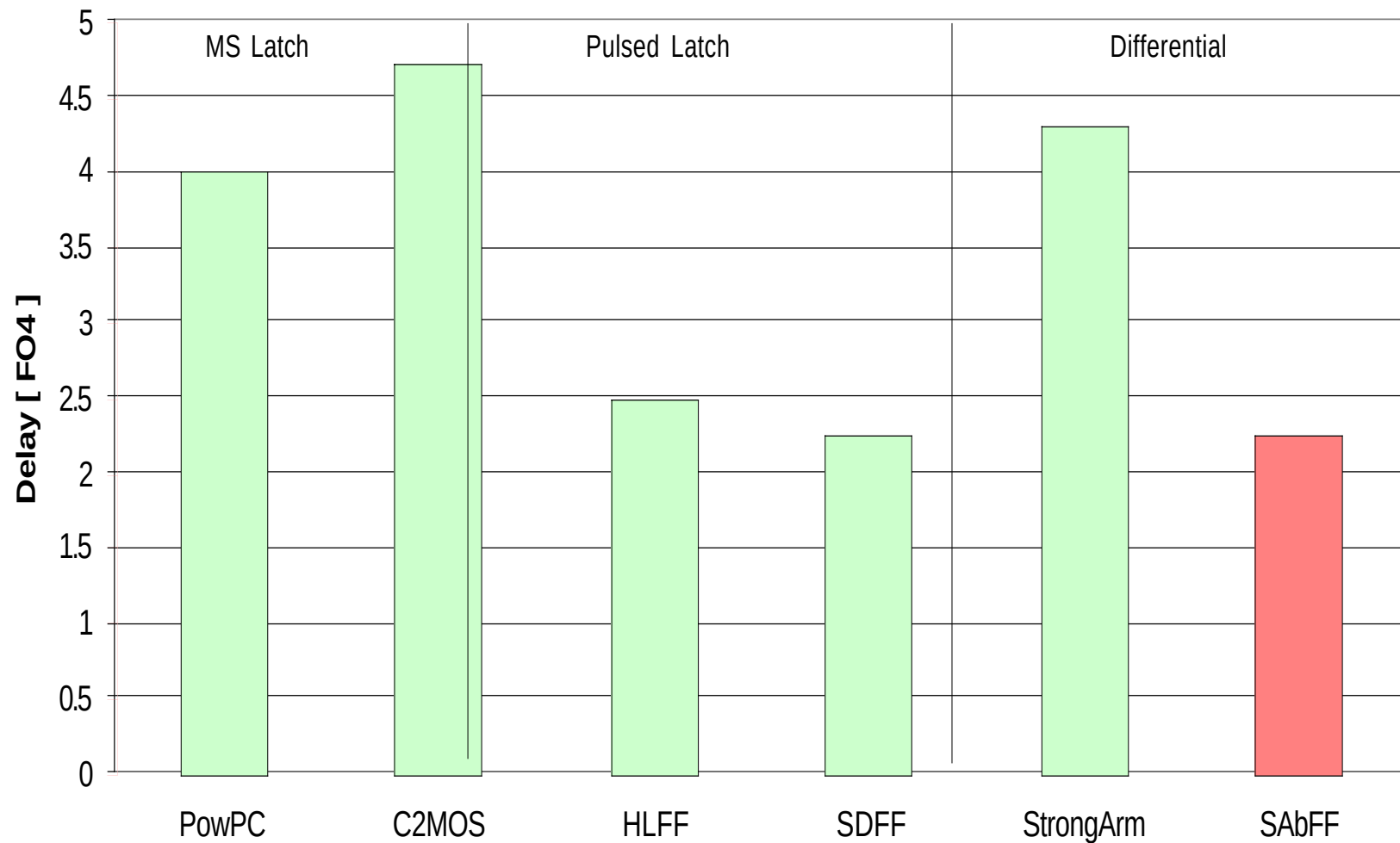
Design & optimization tradeoffs



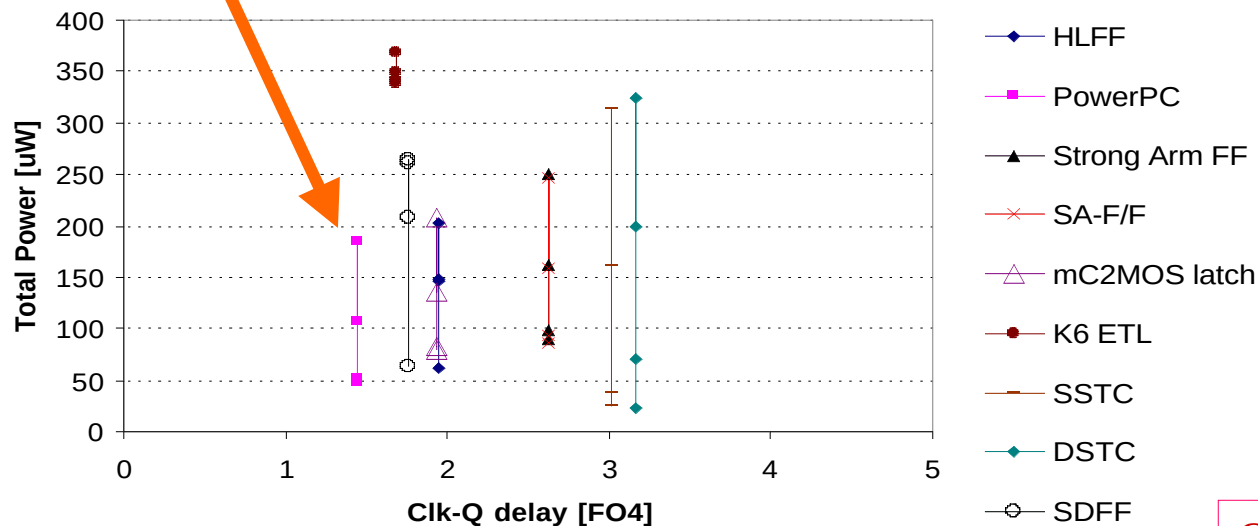
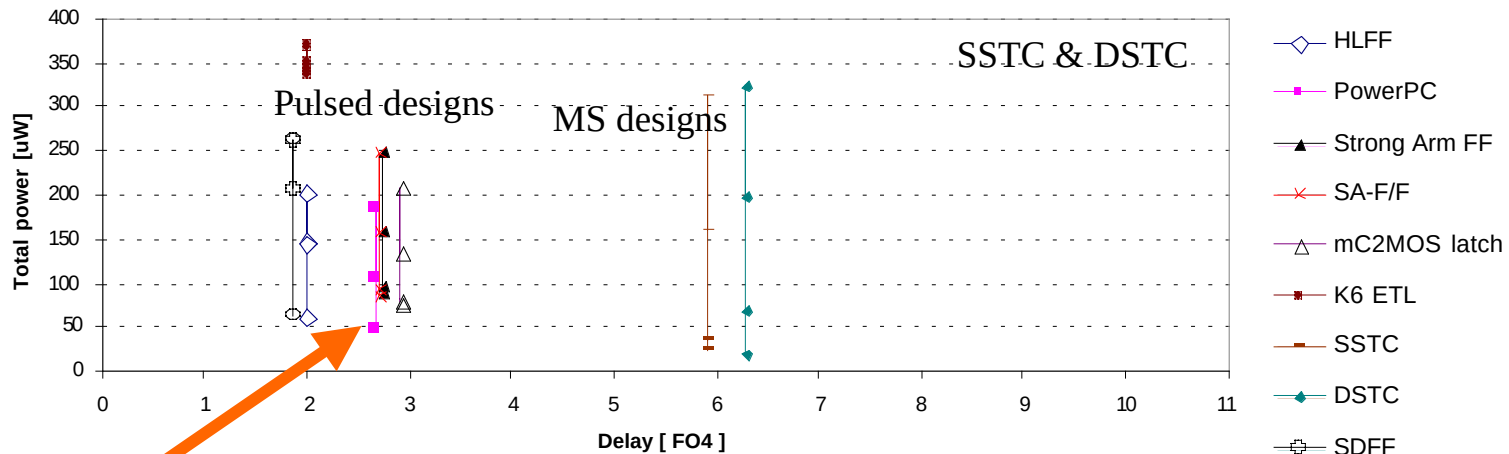
- Opposite Goals
 - Minimal Total power consumption
 - Minimal Delay
- Power-Delay tradeoff
- Minimize Power-Delay product (PDP_{tot}) @ f=const.



Delay Comparison (50% activity)



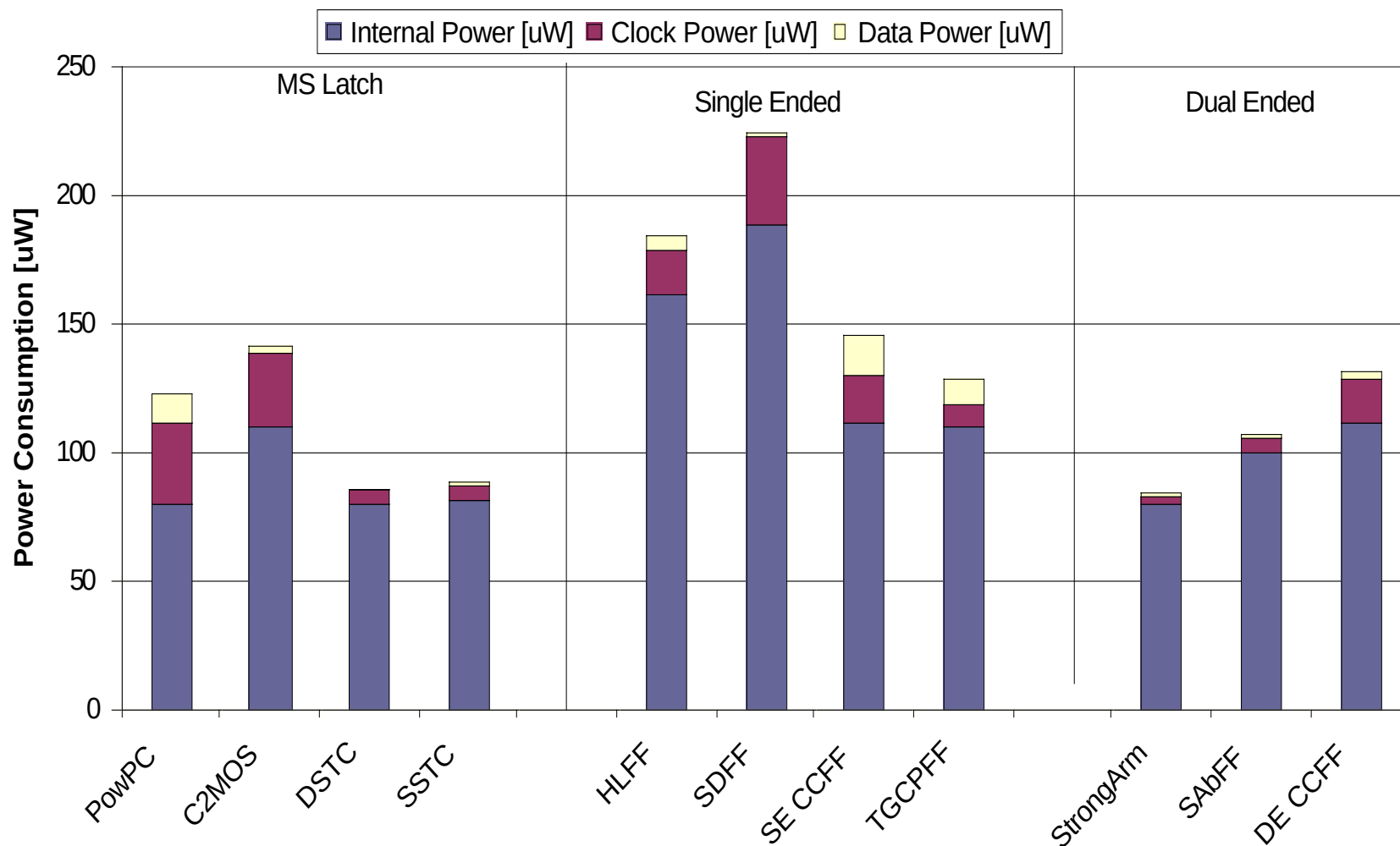
Conventional Clk-Q vs.minimum D-Q



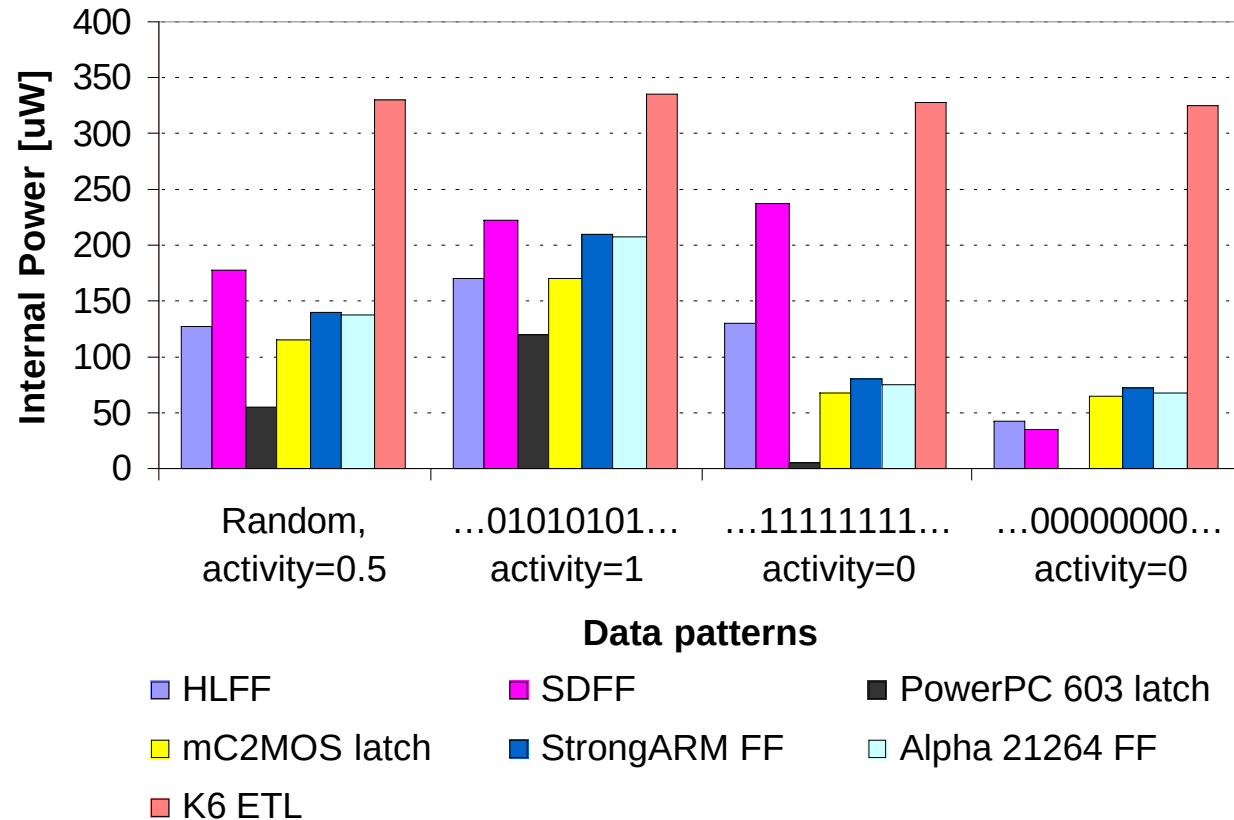
- Hidden positive setup time
- Degradation of total delay

Older 0.22u comparison results

Single-Edge Triggered Structures Power Consumption Comparison (50% activity)



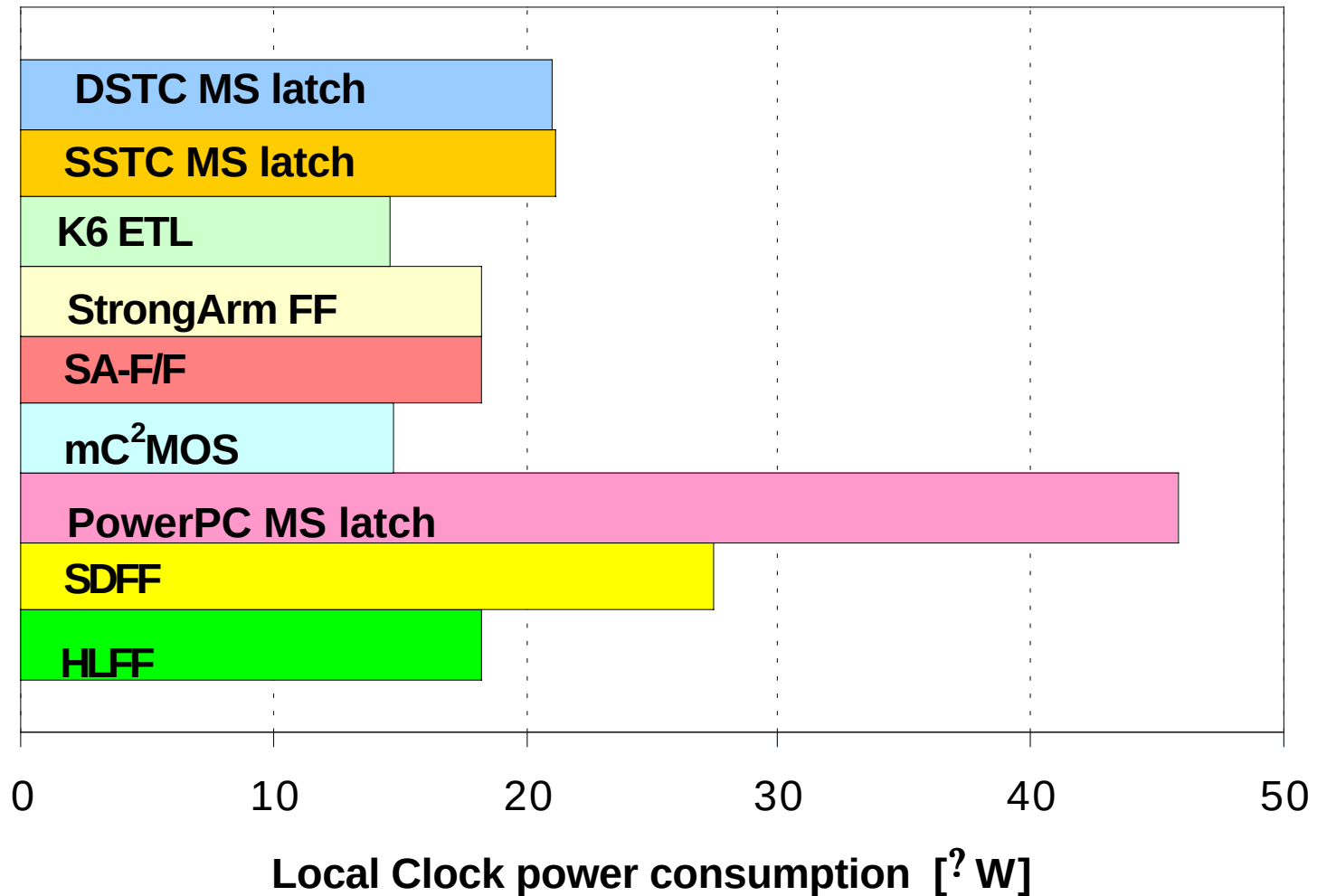
Internal Power distribution



- Four sequences characterize the boundaries for internal power consumption
 - ...010101... maximum
 - random, equal transition probability, average
 - ...111111... precharge activity
 - ...000000... leakage + internal clock processing

Older 0.22u comparison results

Comparison of Clock power consumption



Older 0.22u comparison results

Design goals

- Apply
 - Small clock load
 - Short direct path
 - Reduced node swing
 - Low-power feedback
 - Pulsed design
 - Optimization of both Master and Slave latch
- Avoid
 - Positive setup time
 - Sensitivity to clock slope and skew
 - Dynamic (floating) nodes
 - Dynamic Master latch

Conduct Energy - Delay optimizations

Take into account all sources of power dissipation

ALWAYS use Clk-Q + setup time for max delay

For more details on storage elements check prof. Oklobdzija's ISSCC'02 talk:

<http://www.ece.ucdavis.edu/acsel> *under* Presentations

Simulation Conditions:

- Power Supply Voltage: $V_{DD}=1.8V$ nominal
- Temperature $T=27^{\circ}C$ nominal
- Technology: $0.18\mu m$ Fujitsu
- Fan-Out of 4 Delay = $75pS$
- Transistor Widths
 - Minimal $0.36\mu m$
 - | Maximal $10\mu m$
- Load: 14 minimal inverters in the technology used
- Clock frequency: $500MHz$ ($250MHz$ for Dual-Edge)
- Data/Clock slopes of ideal signal $100ps$